

A Logarithmic Hilbert Compression Gap Dichotomy

Run fa_banach_001, lane 19

June 19, 2026

Source Problem

The source paper is Goulmira Arzhantseva, Cornelia Drutu, and Mark Sapir, *Compression functions of uniform embeddings of groups into Hilbert and Banach spaces*, arXiv:math/0612378. Its Question 5.22 asks:

Theorem 5.5 is proved.

We conclude with some open questions.

Question 5.22. Does every finitely generated group have a Hilbert space compression gap of the form $(\frac{f}{\log x}, f)$ for some function $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$?

In the terminology of Definition 3.3 of the source paper, a pair (f, g) is a Hilbert compression gap for a metric space X if there is an f -embedding of X into Hilbert space and every Hilbert lower compression function ρ satisfies $\rho \ll g$. The relation $a \ll b$ is the usual large-scale comparison: $a(t) \leq Cb(Ct) + C$.

Answer

The literal formulation has a small but important convention issue: a group which does not uniformly embed into Hilbert space cannot have any proper Hilbert compression gap under the strict reading of Definition 3.3. The source paper itself uses logarithmically degraded lower functions in Theorem 5.5, so the result below is stated in that same large-scale compression-gap convention and flags the stricter proper-lower-function reading explicitly.

Theorem 1 (logarithmic annular compression envelope). *Let G be an infinite finitely generated group with a word metric. If G admits a uniform, equivalently coarse, embedding into Hilbert space, then there is a proper nondecreasing function $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that G has, in the source paper's large-scale sense, a Hilbert space compression gap of the form*

$$\left(\frac{f}{\log(x+2)}, f \right)$$

The same construction gives a strict Definition 3.3 compression gap whenever this lower function $f/\log(x+2)$ is proper. Conversely, if a group has a strict proper Hilbert compression gap, then it is Hilbert coarsely embeddable.

Proof Intuition

For each dyadic annulus $2^k \leq d(x, y) < 2^{k+1}$, measure the largest separation that a single 1-Lipschitz Hilbert map can force on every pair in that annulus. Call this number H_k . If the group coarsely embeds, then these annular optima go to infinity.

The upper compression function should not be H_k itself, since annular optima may oscillate. The tail infimum

$$F_k = \inf_{j \geq k} H_j$$

is monotone and still tends to infinity. It is a universal upper bound, up to large-scale comparison, because any lower control at a given scale must also be valid at all larger dyadic scales.

For the lower bound, choose almost optimal maps on all annuli and put them in a Hilbert direct sum with coefficient $1/k$ on the k -th annulus. The coefficients are square summable, so the sum is Lipschitz, while on the k -th annulus the k -th coordinate alone gives the lower bound

$$\frac{H_k}{k} \geq \frac{F_k}{k} \asymp \frac{f}{\log x}.$$

Proof

We use the source paper's convention that a ρ -embedding into Hilbert space is a Lipschitz map $\phi : G \rightarrow \mathcal{H}$ such that

$$\|\phi(x) - \phi(y)\| \geq \rho(d(x, y))$$

for all $x, y \in G$, where ρ is proper and nondecreasing.

Assume that G uniformly embeds into Hilbert space. Fix a word metric d . For each integer $k \geq 1$, define

$$A_k = \{(x, y) \in G^2 : 2^k \leq d(x, y) < 2^{k+1}\}.$$

Let H_k be the supremum, over all 1-Lipschitz maps $\phi : G \rightarrow \mathcal{H}$, of

$$\inf_{(x, y) \in A_k} \|\phi(x) - \phi(y)\|.$$

The numbers H_k are finite, since every 1-Lipschitz map separates pairs in A_k by at most 2^{k+1} .

Because G uniformly embeds into Hilbert space, after rescaling there is a 1-Lipschitz map $\psi : G \rightarrow \mathcal{H}$ and a proper nondecreasing function η such that

$$\|\psi(x) - \psi(y)\| \geq \eta(d(x, y)).$$

Thus $H_k \geq \eta(2^k)$, and $H_k \rightarrow \infty$.

Set

$$F_k = \inf_{j \geq k} H_j.$$

Then F_k is nondecreasing and $F_k \rightarrow \infty$. Define a proper nondecreasing function $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ by setting, for large t ,

$$f(t) = F_k \quad \text{when } 2^k \leq t < 2^{k+1},$$

and extending arbitrarily on bounded intervals.

Universal upper bound

Let ρ be the lower control of any Hilbert ρ -embedding $\phi : G \rightarrow \mathcal{H}$. Suppose ϕ is L -Lipschitz. Then $L^{-1}\phi$ is 1-Lipschitz, so for every j ,

$$L^{-1}\rho(2^j) \leq H_j.$$

If $2^k \leq t < 2^{k+1}$, then for all $j \geq k+1$,

$$\rho(t) \leq \rho(2^j) \leq LH_j.$$

Taking the infimum over $j \geq k+1$ gives

$$\rho(t) \leq LF_{k+1} \leq Lf(2t).$$

Therefore $\rho \ll f$. Since ρ was arbitrary, f is an upper bound for a Hilbert compression gap of G .

Lower bound with one logarithmic loss

For each k , choose a 1-Lipschitz map $\phi_k : G \rightarrow \mathcal{H}_k$ such that

$$\inf_{(x,y) \in A_k} \|\phi_k(x) - \phi_k(y)\| \geq \frac{1}{2}H_k.$$

Fix $e \in G$, replace ϕ_k by $\phi_k - \phi_k(e)$, and define

$$\Phi(x) = \bigoplus_{k=1}^{\infty} \frac{1}{k} \phi_k(x).$$

For each x , this vector is well-defined because $\|\phi_k(x)\| \leq d(x, e)$ and $\sum k^{-2} < \infty$. Moreover,

$$\|\Phi(x) - \Phi(y)\|^2 \leq d(x, y)^2 \sum_{k=1}^{\infty} \frac{1}{k^2},$$

so Φ is Lipschitz.

If $d(x, y) \in [2^k, 2^{k+1})$, then the k -th coordinate gives

$$\|\Phi(x) - \Phi(y)\| \geq \frac{1}{k} \|\phi_k(x) - \phi_k(y)\| \geq \frac{H_k}{2k} \geq \frac{F_k}{2k}.$$

Since $k \asymp \log(d(x, y) + 2)$ and $f(d(x, y)) = F_k$ on this annulus, Φ , after a harmless rescaling, is an

$$\frac{f(x)}{\log(x+2)}$$

embedding. Hence $(f/\log(x+2), f)$ is a Hilbert compression gap.

The functions $\log x$ and $\log(x+2)$ are equivalent for the source paper's large-scale comparison, so this is the requested form.

Finally, if one insists that the lower function in a compression gap be proper, then any group with such a gap is Hilbert coarsely embeddable by definition. Thus non-Hilbert-embeddable finitely generated groups, whose existence is recalled in the source paper's introduction via Gromov's construction, are counterexamples to the literal all-groups reading under that strict convention. $\square$

Verification and Novelty Search

The proof is purely metric and uses only Hilbert direct sums and dyadic annuli. The bounded novelty check used the following searches:

```
Hilbert space compression gap f/log,  
compression gap coarsely embeds Hilbert log,  
optimal compression function coarse embedding Hilbert log,  
arXiv:math/0612378 compression gap.
```

Semantic Scholar metadata for arXiv:math/0612378 showed 35 citations; titles and abstracts containing compression, gap, amenability, wreath products, and Hilbert compression were inspected, with no direct later answer to Question 5.22 found. The result should nevertheless be reviewed for possible folklore status, since the dyadic direct-sum argument is natural.

References

References

- [1] Goul'nara Arzhantseva, Cornelia Drutu, and Mark Sapir, *Compression functions of uniform embeddings of groups into Hilbert and Banach spaces*, arXiv:math/0612378.
- [2] Mikhail Gromov, *Random walk in random groups*, Geom. Funct. Anal. 13 (2003), 73–146.